

Electronics for instrumentation

Instrument: general term that researchers use for a measurement device

Instrumentation: the process of developing, testing, and using the device, or group of instruments used to monitor and control the machine. (Music pieces instruments)

Electronics: circuits or devices using transistors, microchips, and other components.

Electronics for instrumentation: the design, realization and use of electronic systems for the measurement of electrical and non-electrical quantities

Main Objective: Enable you to specify, build and use a set of basic instrumentation and computer based data acquisition systems to acquire data from a laboratory experiment, engineering field site or test vehicle and process it using standard algorithms.

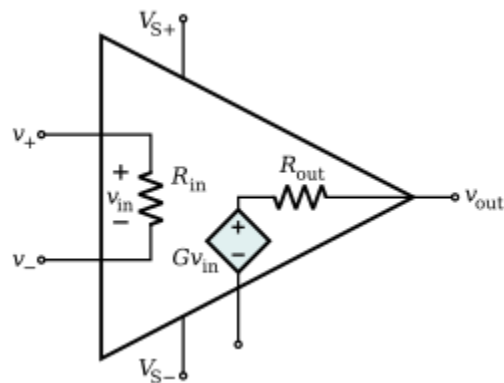
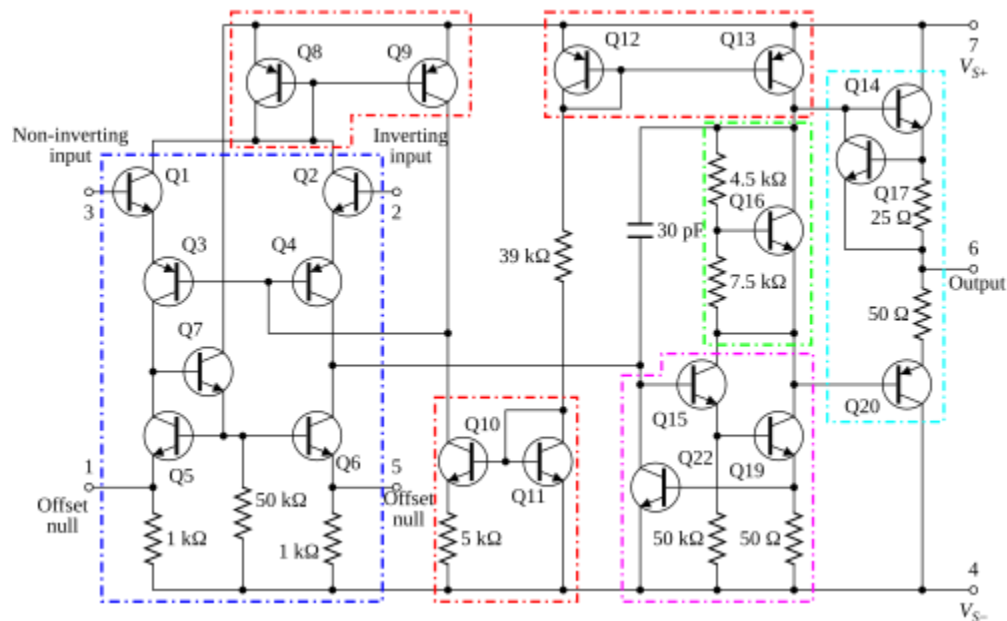
History:

- One of the oldest water clocks was found in the tomb of the ancient Egyptian pharaoh Amenhotep I
- Invented in 1788 by James Watt, this very basic and functional solution was able to measure engine speed through the rotation of two metal balls.
- 1920 electronics instruments were first developed
- 1960s electronic instrumentation was integrated into industry

Importance: Very important as it is one of the most basic steps in creating a controlled system

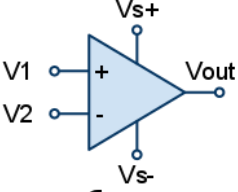
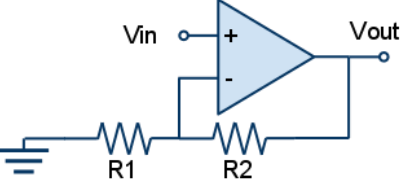
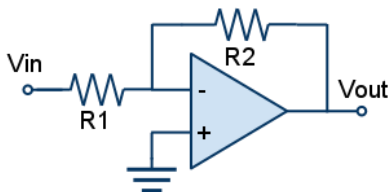
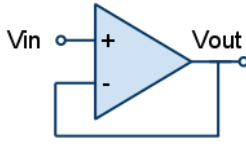
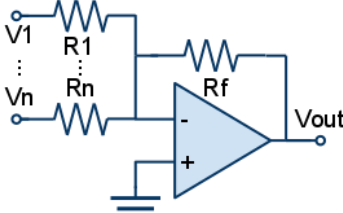
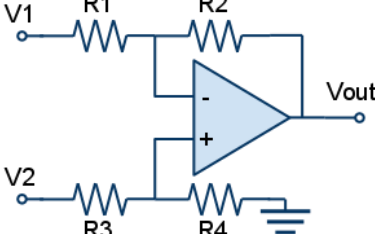
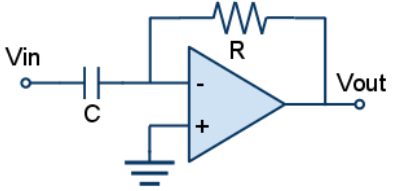
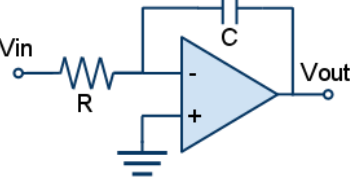
Pre-requisites: Basics of electronics, sensors and measurements, programming

OP-AMP



- It amplifies the difference between the inputs.
- The maximum output voltage is the op-amp supply voltage.
- Infinite input impedance (so that the difference is not changed)
- Zero output impedance (so that all of the output voltage is applied on the load)
- Infinite output voltage range
- Infinite gain
- The op-amp tries to keep the inputs the same voltage

Basic Operational Amplifier Configurations

Voltage Comparator  $V_{out} = \begin{cases} V_{s+} & V1 > V2 \\ V_{s-} & V1 < V2 \end{cases}$	Non-Inverting Amplifier  $V_{out} = V_{in} * \left(1 + \frac{R2}{R1}\right)$	Inverting Amplifier  $V_{out} = -V_{in} * \left(\frac{R2}{R1}\right)$
Voltage Follower  $V_{out} = V_{in}$	Inverting Summing Amplifier  $V_{out} = -R_f * \left(\frac{V1}{R1} + \dots + \frac{Vn}{Rn}\right)$	Differential Amplifier  $V_{out} = \left(1 + \frac{R2}{R1}\right) \left(\frac{R4}{R3 + R4}\right) * V2 - \left(\frac{R2}{R1}\right) * V1$ If $R1 = R3$ and $R2 = R4$ Then $V_{out} = \left(\frac{R2}{R1}\right) (V2 - V1)$
Differentiator Amplifier  $V_{out} = -R * C * \left(\frac{dV_{in}}{dt}\right)$	Integrator Amplifier  $V_{out} = -\left(\frac{1}{R * C}\right) V_{in} dt$	